

Reto 1 Azure Monitor Budget Alert y

Una alerta de presupuesto monitorea el gasto acumulado del Resource Group y notifica por email cuando supera un umbral definido. Esto es esencial en suscripciones con crédito limitado como Azure for Students, donde un recurso dejado encendido puede agotar el crédito sin que el equipo se dé cuenta.

Implementación:

Se agregó el recurso `azurerm_consumption_budget_resource_group` al final de `infra/main.tf`. Este recurso crea una alerta mensual de \$10 USD que notifica al correo del equipo cuando el gasto supera el 80%:

```
resource "azurerm_consumption_budget_resource_group" "budget" {
  name           = "${var.prefix}-budget"
  resource_group_id = azurerm_resource_group.rg.id
  amount         = 10
  time_grain      = "Monthly"

  time_period {
    start_date = "2026-03-01T00:00:00Z"
  }

  notification {
    enabled      = true
    threshold     = 80
    operator      = "GreaterThan"
    contact_emails = ["robinson.nunez-p@mail.escuelaing.edu.co"]
  }
}
```

Parámetros configurados

- `amount = 10`: Presupuesto mensual de \$10 USD
- `time_grain = Monthly`: Se evalúa mes a mes
- `threshold = 80`: Alerta cuando el gasto supera el 80% \$8 USD
- `operator = GreaterThan`: Dispara cuando el gasto real supera el umbral
- `contact_emails`: Notificación al correo del equipo de desarrollo

Azure Monitor Metric Alert sobre el health probe:

Una alerta de métrica monitorea la disponibilidad de las VMs en el backend pool del Load Balancer. La métrica `DipAvailability` indica qué porcentaje de las instancias están respondiendo correctamente al health probe TCP/80. Si alguna VM cae, el valor baja de 100 y se dispara la alerta.

Paso 1 Exponer el ID del Load Balancer

Para que el recurso de alerta pueda referenciar el Load Balancer, primero se agregó un output `lb_id` al módulo `lb` en `modules/lb/main.tf`:

```
output "lb_id" {
  value = azurerm_lb.lb.id
}
```

Paso 2 Crear el recurso de alerta

Se agregó el recurso `azurerm_monitor_metric_alert` al final de `infra/main.tf`. La alerta se activa cuando la disponibilidad promedio del health probe cae por debajo del 100% en una ventana de 15 minutos:

```
resource "azurerm_monitor_metric_alert" "probe_alert" {
  name                       = "${var.prefix}-probe-alert"
  resource_group_name       = azurerm_resource_group.rg.name
  scopes                    = [module.lb.lb_id]
  severity                  = 2
  frequency                 = "PT5M"
  window_size               = "PT15M"

  criteria {
    metric_namespace = "Microsoft.Network/loadBalancers"
    metric_name      = "DipAvailability"
    aggregation      = "Average"
    operator         = "LessThan"
    threshold        = 100
  }

  tags = var.tags
}
```

Validación y despliegue

Se ejecutó `terraform validate` para confirmar que los nuevos recursos son sintácticamente correctos, seguido de `terraform apply` para crearlos en Azure:

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> terraform validate
Success! The configuration is valid.

PS C:\Users\robin\downloads\LAB08_ARSW\infra> terraform plan -var-file env/dev.tfvars -out plan.tfplan
Acquiring state lock. This may take a few moments...
azurerm_resource_group.rg: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg]
module.lb.azure_rm_public_ip.pip: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/publicIPAddresses/lab8-lb-pip]
module.vnet.azure_rm_virtual_network.vnet: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/virtualNetworks/lab8-vnet]
module.lb.azure_rm_network_security_group.nsg: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/networkSecurityGroups/lab8-web-nsg]
module.lb.azure_rm_lb.lb: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/loadBalancers/lab8-lb]
module.vnet.azure_rm_subnet.mgmt: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/virtualNetworks/lab8-vnet/subnets/subnet-mgmt]
module.vnet.azure_rm_subnet.web: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/virtualNetworks/lab8-vnet/subnets/subnet-web]
module.lb.azure_rm_lb_probe.probe: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/
```

Resultado del apply

Terraform creó los 2 recursos nuevos sin tocar ninguno de los 18 recursos existentes (VMs, LB, VNet, NSG). El budget alert se creó en 7 segundos y el monitor alert tardó 2 minutos por la propagación en Azure Monitor:

```
PS C:\Users\robin\downloads\LAB08_ARSW\infra> terraform apply -var-file env/dev.tfvars -var "ssh_public_key=ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIAamJIEy9GPmSe7J/4P2Nc6t3hikRvoppZTSQNW0N1 robinson.nunez-p@mail.escuelaing.edu.co"
Acquiring state lock. This may take a few moments...
azurerm_resource_group.rg: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg]
module.lb.azure_rm_public_ip.pip: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/publicIPAddresses/lab8-lb-pip]
module.vnet.azure_rm_virtual_network.vnet: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/virtualNetworks/lab8-vnet]
module.lb.azure_rm_network_security_group.nsg: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/networkSecurityGroups/lab8-web-nsg]
module.lb.azure_rm_lb.lb: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/loadBalancers/lab8-lb]
module.vnet.azure_rm_subnet.mgmt: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/virtualNetworks/lab8-vnet/subnets/subnet-mgmt]
module.vnet.azure_rm_subnet.web: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/virtualNetworks/lab8-vnet/subnets/subnet-web]
module.compute.azure_rm_network_interface.nic[1]: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/networkInterfaces/lab8-nic-1]
module.compute.azure_rm_network_interface.nic[0]: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/networkInterfaces/lab8-nic-0]
module.lb.azure_rm_backend_address_pool.bepool: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Network/loadBalancers/lab8-lb/backendAddressPools/lab8-bepool]
module.lb.azure_rm_lb_probe.probe: Refreshing state... [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/

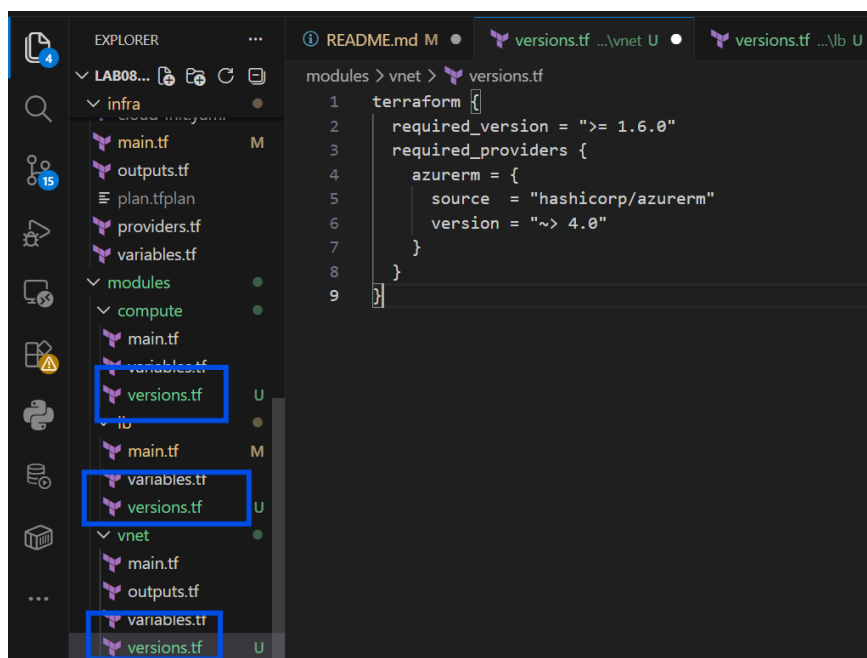
azurerm_monitor_metric_alert.probe.alert: Creating...
azurerm_consumption_budget_resource_group.budget: Creating...
azurerm_consumption_budget_resource_group.budget: Creation complete after 7s [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Consumption/budgets/lab8-budget]
azurerm_monitor_metric_alert.probe.alert: Still creating... [00m10s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [00m20s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [00m30s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [00m40s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [00m50s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [01m00s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [01m10s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [01m20s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [01m30s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [01m40s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [01m50s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [02m00s elapsed]
azurerm_monitor_metric_alert.probe.alert: Still creating... [02m10s elapsed]
azurerm_monitor_metric_alert.probe.alert: Creation complete after 2m11s [id=/subscriptions/056e8de1-7fc3-4ecb-ad1d-1a1255a82e7a/resourceGroups/lab8-rg/providers/Microsoft.Insights/metricAlerts/lab8-probe-alert]
Releasing state lock. This may take a few moments...

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.

Outputs:
lb_public_ip = "20.114.114.201"
resource_group_name = "lab8-rg"
vm_names = [
  "lab8-vm-0",
  "lab8-vm-1",
]
PS C:\Users\robin\downloads\LAB08_ARSW\infra>
```

RETO 2 Módulos privados versionados con semantic versioning:

1. Agrega un archivo versions.tf a cada módulo modules/vnet/, modules/compute/ y modules/lb/.



Validacion del TAG:

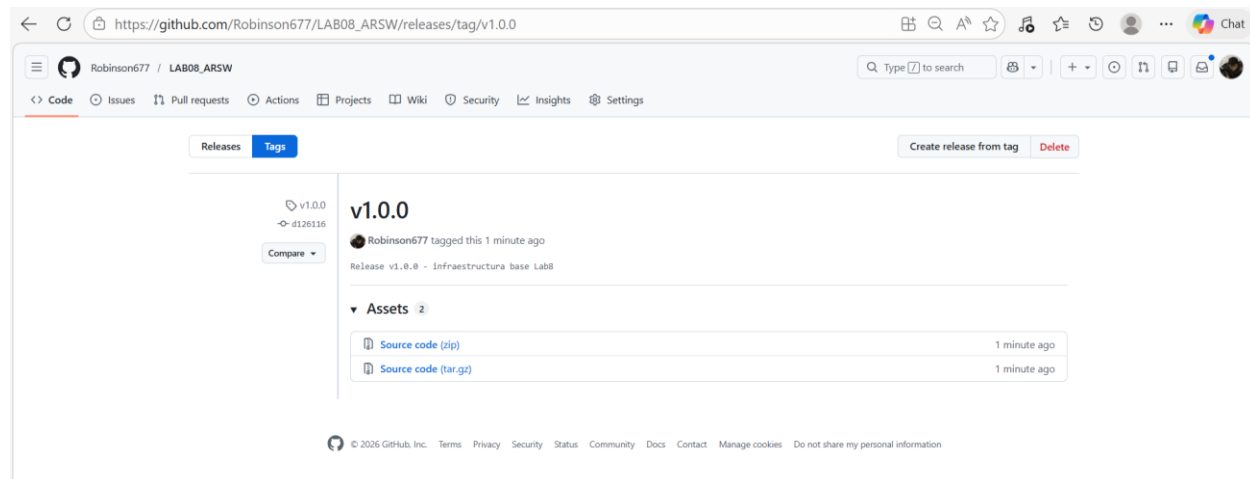
- Creamos el TAG

```
robin@Kawaki MINGW64 ~/downloads/LAB08_ARSW (main)
$ git tag -a v1.0.0 -m "Release v1.0.0 - infraestructura base Lab8"

robin@Kawaki MINGW64 ~/downloads/LAB08_ARSW (main)
$ git push origin v1.0.0
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 202 bytes | 202.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Robinson677/LAB08_ARSW.git
 * [new tag]           v1.0.0 -> v1.0.0

robin@Kawaki MINGW64 ~/downloads/LAB08_ARSW (main)
```

Luego verificamos en https://github.com/Robinson677/LAB08_ARSW/tags



Volvemos a validar

